

Hey, everybody. Welcome to your chapter 9 review. In this episode, we're going to review nutrition and talk about some of the key learning points from this chapter. All right. Admit it. How many of you have tried a trendy diet that had a really good marketing and came with a free tote bag and a super cool water bottle? You can't see it, but I'm raising my hand right now. And whether you raised your hand or not, I know I'm not alone. It seems like there's something new every day. Some are backed by solid scientific research, but others only have the benefit of free merchandise.

Ultimately, it'll be your responsibility to be the filter for the abundance of nutrition information that's available nowadays. As a fitness professional, it'll be important to help your clients distinguish between reliable, science-based nutrition versus popular trends and fads. Depending on the goal, nutrition is arguably as important, if not maybe more important, than exercise as a vital component of a client's overall program.

Nutrient intake directly impacts how a body will respond to an exercise program. But no matter how hard somebody tries, they simply won't be able to out-exercise a bad diet. At a basic level, nutrition is defined as the process of acquiring nutrients through food to support energy needs and cellular processes like growth, repair, and maintenance of tissues, reproduction, digestion, and respiration. When there are excesses or deficiencies in nutrition, it can have profound effects on overall development and health.

Nutrition also plays a key role in optimizing exercise performance and physical activities as well as modifying body composition, such as the increase of lean muscle mass or the loss of body fat. An understanding of basic nutrition principles is needed to properly guide your clients on the importance of good nutrition practices to complement and enhance their training program. Without proper nutrition, their training could be greatly compromised, regardless of whether it's geared toward fat loss, muscle gain, or sports performance.

If good stuff isn't going into the body, good results aren't going to come out of it. Let's give you the lowdown on specific goals. By the end of our chapter on nutrition, you need to be able to explain the role of macronutrients, micronutrients, and hydration in achieving and maintaining optimal health. Compare scope of practice of allied health professionals and certified personal trainers when providing nutrition education. Determine the reliability, validity, and credibility of nutrition information from various source types.

Communicate information to clients with differing knowledge levels of nutrition. And identify strategies that empower clients to make nutritional decisions affecting body composition. The first point I want to make is that, yes, nutrition plays an important role in health and exercise. And a basic understanding of the subject can help you better meet the needs of your clients as they move toward their performance or training goals.

However, it's incredibly important to keep scope of practice in mind. Providing nutrition counseling, medical nutrition therapy, and specific meal plans are not within the scope of practice of fitness professionals who don't hold any other nutrition credentials, like a licensed dietitian. You can absolutely provide general guidelines on good nutrition practices and explain the importance of nutrition and hydration for health, training, and performance.

However, you should direct your clients to a credible, registered, and licensed dietitian to provide nutrition counseling, medical nutrition therapy, and meal plans as needed. But let's assume you've taken scope of practice into consideration. There is a ton you can do to help your clients with nutrition. Something as simple as educating clients on food packaging can make a huge impact. Food labels give us important information on ingredients and nutritional values and can help clients make informed decisions about how a food item contributes to their nutrition and fitness goals.

Knowing how to dissect the nutrition facts panel and the ingredients list enables clients to evaluate the composition and nutritional value of food products in relation to their goals. Educating clients on ways to use food labels can be an important component of a nutrition strategy for body composition and performance goals. Having this understanding can help your clients feel empowered to make better choices.

Let's look more closely at the different types of macronutrients that provide usable energy by cells for various processes. Protein is one of the three macronutrients. It is comprised of 20 amino acids, 9 of which are considered essential, meaning they must be obtained through dietary intake. The other 11, which are considered non-essential, can be made in the body as long as there are adequate levels of essential amino acids available.

The role of protein is the synthesis and repair of tissues, organs, hormones, enzymes, and peptides. Dietary sources of complete proteins include soy and animal foods, like meat, poultry, seafood, and dairy. Plant-based incomplete protein foods include legumes, grains, and vegetables.

The current Recommended Dietary Allowance, or RDA, for protein differs depending on the individual. It's going to be greater for athletes and most exercising individuals due to increased protein turnover from muscle breakdown and recovery as well as an increase in oxidation of amino acids to meet energy needs. But it's typically in the range of 10% to 35% of total calories.

Moving on to carbohydrates, or carbs, as we lovingly refer to them, they are important for optimal training and sports performance and exist in several different forms-- glycogen, simple sugars, complex carbohydrates, and fiber. Glycogen is the storage form of carbohydrates and is stored in the liver and skeletal muscle. Simple sugars include the monosaccharides-- glucose, fructose, galactose-- and disaccharides-- lactose, sucrose, maltose.

Complex carbohydrates are long chains of glucose units called polysaccharides, which are slower to digest and raise blood glucose levels slowly. Your dietary carb sources include plant foods and dairy, like grains, vegetables, legumes, fruit, milk, and yogurt. Fiber is indigestible carbohydrate associated with various health benefits and includes both soluble and insoluble fiber. With the exception of fiber, carbohydrates provide 4 calories per gram and are considered an important source of fuel for the body, especially for those who exercise regularly.

The acceptable macronutrient distribution range for carbs is 45% to 65% of calories in the diet. The third and final category of macronutrients are lipids, commonly referred to as fats. Lipids include triglycerides, phospholipids, and sterols, containing 9 calories per gram. They are a concentrated but important source of energy for active individuals.

Some types of lipids, such as monounsaturated fats and polyunsaturated omega-3 fats, are associated with health benefits like reduced inflammation and improved heart health. Other lipids, like manufactured trans fats, can contribute to adverse health effects when consumed in excess. Dietary sources of lipids include solid fats and oils in foods like meats, fish, poultry, dairy, egg yolks, legumes, nuts, and seeds. Current dietary guidelines suggest the majority of your lipid intake should come in the form of heart-healthy sources, like fish, legumes, nuts, and seeds, to make up 20% to 35% of total calories.

And now onto the micronutrients-- these include vitamins, minerals, and phytonutrients or plant-based nutrients. They are not considered energy sources, but they are important in the regulation of various metabolic processes, including energy metabolism. Vitamin and mineral deficiencies and insufficiencies can also contribute to various health issues.

Vitamins include two groups, fat soluble and water soluble. The fat-soluble vitamins include vitamins A, D, E, and K. The water-soluble vitamins consist of vitamin C and a large group of B vitamins. Minerals are compounds that you'll find in both plant and animal foods as well as the environment. They're required in the diet to support various health and metabolic processes. Major minerals are present in the body in larger amounts and are also required in larger quantities in the diet as compared to trace minerals.

Having a well-balanced, healthy diet comprised mostly of minimally processed whole foods is a good source for micronutrients. However, supplementation might be helpful and necessary in cases of inadequate intakes, deficiencies, and/or increased nutrient needs, such as injury, illness, and high-volume training. In these cases, it's a good idea to have clients work with a medical professional to diagnose any suspected issues.

Water intake is a major factor of nutrition and wellness. Hydration serves many important functions in the human body, including the regulation of body temperature, transport and distribution of water-soluble nutrients, maintenance of blood volume, lubrication of joints, membranes, and synovial tissue, shock absorption, and removal of waste matter and toxins. Adequate hydration is essential for optimal health, training, and performance.

If a person isn't getting enough fluids, it can affect physiologic functions and cognitive processes. Dehydration can also impede athletic performance and incur a variety of negative effects on the body, like increased heart rate, higher body temperature, and decreased blood flow, to name a few. Athletes have even higher hydration needs and should replace fluid at 1.25 times the amount of bodyweight lost during an event.

Sports drinks are a popular go-to source for hydration but likely unnecessary for short-duration exercise lasting less than 60 minutes, unless in hot or humid temperatures. In that case, drink up and hope someone also dumps an ice-cold cooler of the stuff over your head. It makes a great photo op. Speaking of which, as you've probably seen, we are inundated with advertising and nutritional studies. I'm saying studies in air quotes, by the way.

We see them on social media, TV and in different publications. But it's important to remember that nutrition is a scientific field with ongoing research. And we have ways of verifying the credibility of information. Consider the source of the information, what qualifications are held by the individuals providing it, and whether it was supported by additional peer-reviewed research. Clients may not always be able to distinguish between reliable, credible nutrition versus popular trends and fads.

As a fitness pro, you can bridge the gap by understanding where to find scientific based information and guiding clients to these sources for further nutrition information. Now, over the course of your career as a fitness pro, you'll be asked to help with weight management, both from the standpoint of losing weight and gaining weight. This will include nutrition strategies in addition to an exercise program.

Losses and gains will focus on a combination of factors, including energy intake, physical activity, sleep, hormones, medications, and genetics. It's great to have a basic understanding of the influence of these factors as they relate to weight management and performance. Your initial assessment with clients is a great time to ask them about all of those factors and see if they might be affecting their goals.

And look, we all know that total energy intake and expenditure are the foundation for weight management. But there is a lot of nuance based on the actual goal. Many clients will have a weight management or performance goal, whether it's fat loss, muscle gain, or improved sport performance. Achievement of these goals relies on an effective training program and appropriate nutrition strategies that complement each other.

For instance, fat loss requires a deficit in caloric intake while minimizing the loss of lean body mass and reduction in Total Daily Energy Expenditure, or TDEE. Some strategies to achieve a net calorie deficit include a moderate reduction in caloric intake and/or an increase in energy expenditure. However, building muscle and enhancing Fat Free Mass, or FFM, is different. It's an anabolic process that requires both the stimulation of Muscle Protein Synthesis, or MPS, and the availability of adequate fuel.

Muscle gain strategies will include resistance training combined with adequate caloric and protein intake, with an emphasis on high biological sources of protein that provide all of the essential amino acids, and potentially a caloric surplus in the range of 300 to 500 calories per day. I guess what I'm getting at is that every client will have different goals, needs, histories, and challenges. The more you know about nutrition, the more effectively you can put that knowledge to creative use, designing smart, science-based suggestions to help each individual reach their own unique goals.

So go ahead and chew on all of that for a while, pun very much intended. OK. Be well, and happy studying.